

Basic Differentiation Rules

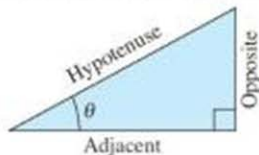
- $\frac{d}{dx}[cu] = cu'$
- $\frac{d}{dx}[u \pm v] = u' \pm v'$
- $\frac{d}{dx}[uv] = uv' + vu'$
- $\frac{d}{dx}\left[\frac{u}{v}\right] = \frac{vu' - uv'}{v^2}$
- $\frac{d}{dx}[c] = 0$
- $\frac{d}{dx}[u^n] = nu^{n-1}u'$
- $\frac{d}{dx}[x] = 1$
- $\frac{d}{dx}[|u|] = \frac{u}{|u|}(u'), \quad u \neq 0$
- $\frac{d}{dx}[e^u] = e^u u'$
- $\frac{d}{dx}[\log_a u] = \frac{u'}{(\ln a)u}$
- $\frac{d}{dx}[a^u] = (\ln a)a^u u'$
- $\frac{d}{dx}[\sin u] = (\cos u)u'$
- $\frac{d}{dx}[\cos u] = -(\sin u)u'$
- $\frac{d}{dx}[\tan u] = (\sec^2 u)u'$
- $\frac{d}{dx}[\cot u] = -(\csc^2 u)u'$
- $\frac{d}{dx}[\csc u] = -(\csc u \cot u)u'$
- $\frac{d}{dx}[\arcsin u] = \frac{u'}{\sqrt{1-u^2}}$
- $\frac{d}{dx}[\arccos u] = \frac{-u'}{\sqrt{1-u^2}}$
- $\frac{d}{dx}[\arctan u] = \frac{u'}{1+u^2}$
- $\frac{d}{dx}[\operatorname{arccot} u] = \frac{-u'}{1+u^2}$
- $\frac{d}{dx}[\operatorname{arcsec} u] = \frac{u'}{|u|\sqrt{u^2-1}}$
- $\frac{d}{dx}[\operatorname{arccsc} u] = \frac{-u'}{|u|\sqrt{u^2-1}}$
- $\frac{d}{dx}[\sinh u] = (\cosh u)u'$
- $\frac{d}{dx}[\cosh u] = (\sinh u)u'$
- $\frac{d}{dx}[\tanh u] = (\operatorname{sech}^2 u)u'$
- $\frac{d}{dx}[\coth u] = -(\operatorname{csch}^2 u)u'$
- $\frac{d}{dx}[\operatorname{csch} u] = -(\operatorname{csch} u \coth u)u'$
- $\frac{d}{dx}[\sinh^{-1} u] = \frac{u'}{\sqrt{u^2+1}}$
- $\frac{d}{dx}[\cosh^{-1} u] = \frac{u'}{\sqrt{u^2-1}}$
- $\frac{d}{dx}[\tanh^{-1} u] = \frac{u'}{1-u^2}$
- $\frac{d}{dx}[\coth^{-1} u] = \frac{u'}{1-u^2}$
- $\frac{d}{dx}[\operatorname{sech}^{-1} u] = \frac{-u'}{u\sqrt{1-u^2}}$
- $\frac{d}{dx}[\operatorname{csch}^{-1} u] = \frac{-u'}{|u|\sqrt{1+u^2}}$

Basic Integration Formulas

- $\int kf(u) du = k \int f(u) du$
- $\int [f(u) \pm g(u)] du = \int f(u) du \pm \int g(u) du$
- $\int du = u + C$
- $\int a^u du = \left(\frac{1}{\ln a}\right)a^u + C$
- $\int e^u du = e^u + C$
- $\int \sin u du = -\cos u + C$
- $\int \cos u du = \sin u + C$
- $\int \tan u du = -\ln|\cos u| + C$
- $\int \cot u du = \ln|\sin u| + C$
- $\int \sec u du = \ln|\sec u + \tan u| + C$
- $\int \csc u du = -\ln|\csc u + \cot u| + C$
- $\int \sec^2 u du = \tan u + C$
- $\int \csc^2 u du = -\cot u + C$
- $\int \sec u \tan u du = \sec u + C$
- $\int \csc u \cot u du = -\csc u + C$
- $\int \frac{du}{\sqrt{a^2-u^2}} = \arcsin \frac{u}{a} + C$
- $\int \frac{du}{u\sqrt{u^2-a^2}} = \frac{1}{a} \operatorname{arcsec} \frac{|u|}{a} + C$

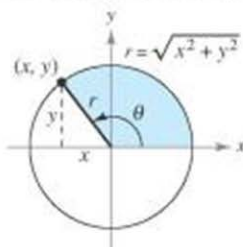
Definition of the Six Trigonometric Functions

Right triangle definitions, where $0 < \theta < \pi/2$.

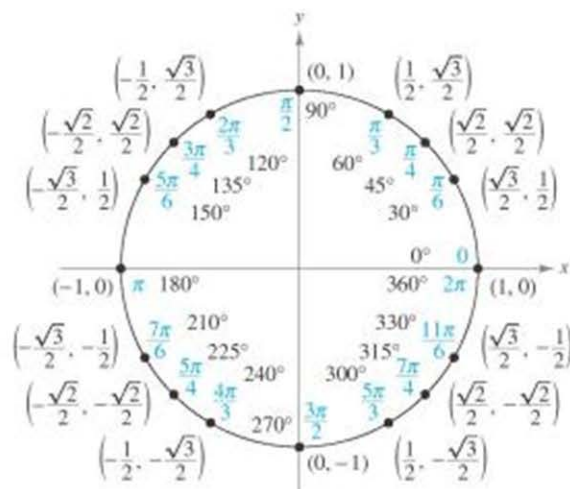


$$\begin{aligned}\sin \theta &= \frac{\text{opp}}{\text{hyp}} & \csc \theta &= \frac{\text{hyp}}{\text{opp}} \\ \cos \theta &= \frac{\text{adj}}{\text{hyp}} & \sec \theta &= \frac{\text{hyp}}{\text{adj}} \\ \tan \theta &= \frac{\text{opp}}{\text{adj}} & \cot \theta &= \frac{\text{adj}}{\text{opp}}\end{aligned}$$

Circular function definitions, where θ is any angle.



$$\begin{aligned}\sin \theta &= \frac{y}{r} & \csc \theta &= \frac{r}{y} \\ \cos \theta &= \frac{x}{r} & \sec \theta &= \frac{r}{x} \\ \tan \theta &= \frac{y}{x} & \cot \theta &= \frac{x}{y}\end{aligned}$$



Reciprocal Identities

$$\begin{aligned}\sin x &= \frac{1}{\csc x} & \sec x &= \frac{1}{\cos x} & \tan x &= \frac{1}{\cot x} \\ \csc x &= \frac{1}{\sin x} & \cos x &= \frac{1}{\sec x} & \cot x &= \frac{1}{\tan x}\end{aligned}$$

Tangent and Cotangent Identities

$$\tan x = \frac{\sin x}{\cos x} \quad \cot x = \frac{\cos x}{\sin x}$$

Pythagorean Identities

$$\begin{aligned}\sin^2 x + \cos^2 x &= 1 \\ 1 + \tan^2 x &= \sec^2 x & 1 + \cot^2 x &= \csc^2 x\end{aligned}$$

Cofunction Identities

$$\begin{aligned}\sin\left(\frac{\pi}{2} - x\right) &= \cos x & \cos\left(\frac{\pi}{2} - x\right) &= \sin x \\ \csc\left(\frac{\pi}{2} - x\right) &= \sec x & \tan\left(\frac{\pi}{2} - x\right) &= \cot x \\ \sec\left(\frac{\pi}{2} - x\right) &= \csc x & \cot\left(\frac{\pi}{2} - x\right) &= \tan x\end{aligned}$$

Reduction Formulas

$$\begin{aligned}\sin(-x) &= -\sin x & \cos(-x) &= \cos x \\ \csc(-x) &= -\csc x & \tan(-x) &= -\tan x \\ \sec(-x) &= \sec x & \cot(-x) &= -\cot x\end{aligned}$$

Sum and Difference Formulas

$$\begin{aligned}\sin(u \pm v) &= \sin u \cos v \pm \cos u \sin v \\ \cos(u \pm v) &= \cos u \cos v \mp \sin u \sin v \\ \tan(u \pm v) &= \frac{\tan u \pm \tan v}{1 \mp \tan u \tan v}\end{aligned}$$

Double-Angle Formulas

$$\begin{aligned}\sin 2u &= 2 \sin u \cos u \\ \cos 2u &= \cos^2 u - \sin^2 u = 2 \cos^2 u - 1 = 1 - 2 \sin^2 u \\ \tan 2u &= \frac{2 \tan u}{1 - \tan^2 u}\end{aligned}$$

Power-Reducing Formulas

$$\begin{aligned}\sin^2 u &= \frac{1 - \cos 2u}{2} \\ \cos^2 u &= \frac{1 + \cos 2u}{2} \\ \tan^2 u &= \frac{1 - \cos 2u}{1 + \cos 2u}\end{aligned}$$

Sum-to-Product Formulas

$$\begin{aligned}\sin u + \sin v &= 2 \sin\left(\frac{u+v}{2}\right) \cos\left(\frac{u-v}{2}\right) \\ \sin u - \sin v &= 2 \cos\left(\frac{u+v}{2}\right) \sin\left(\frac{u-v}{2}\right) \\ \cos u + \cos v &= 2 \cos\left(\frac{u+v}{2}\right) \cos\left(\frac{u-v}{2}\right) \\ \cos u - \cos v &= -2 \sin\left(\frac{u+v}{2}\right) \sin\left(\frac{u-v}{2}\right)\end{aligned}$$

Product-to-Sum Formulas

$$\begin{aligned}\sin u \sin v &= \frac{1}{2} [\cos(u-v) - \cos(u+v)] \\ \cos u \cos v &= \frac{1}{2} [\cos(u-v) + \cos(u+v)] \\ \sin u \cos v &= \frac{1}{2} [\sin(u+v) + \sin(u-v)] \\ \cos u \sin v &= \frac{1}{2} [\sin(u+v) - \sin(u-v)]\end{aligned}$$